Combining Geometric Transformations

Introduction
Determining the Image of a Point Given a Sequence of Transformations
Applying Multiple Transformations to a Point Using Functions
Determining the Image of a Polygon Under the Action of Multiple Transformations
Determining Parameters Given a Point and Its Image Under Multiple Transformations

Prerequisites

- Dilations of Figures in the Coordinate Plane
- Combining Stretches of Geometric Figures
- Rotating Objects in the Coordinate Plane Using Functions

Introduction

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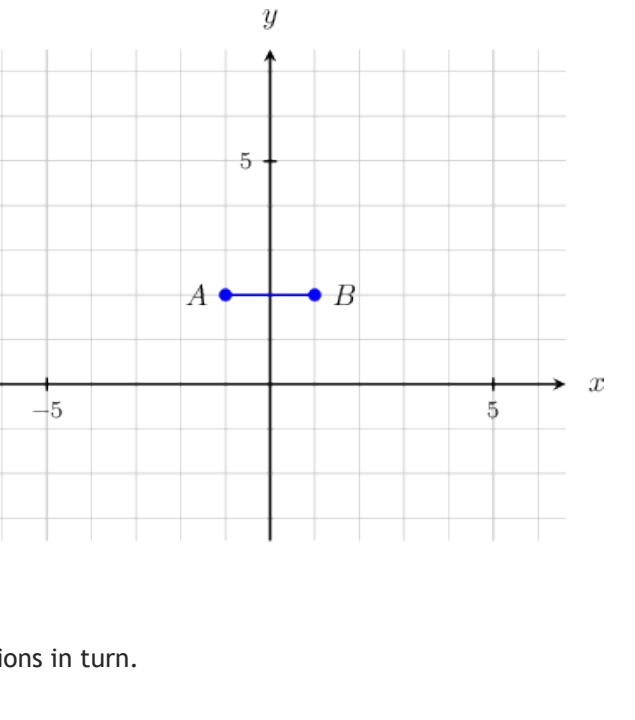
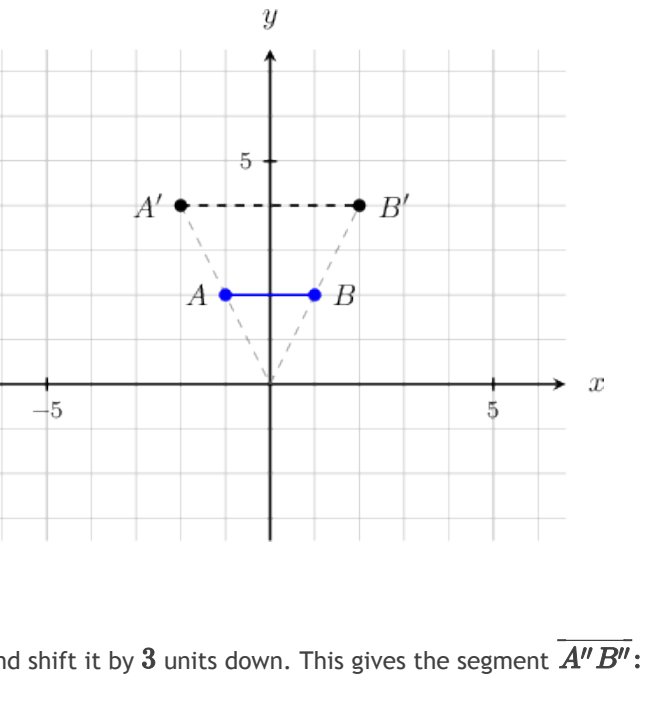
We can combine transformations by applying them in turn.

For example, suppose that the segment $\overline{AB}$ below is dilated with center O and a scale factor 2, and then shifted 3 units down. What is the resulting segment?

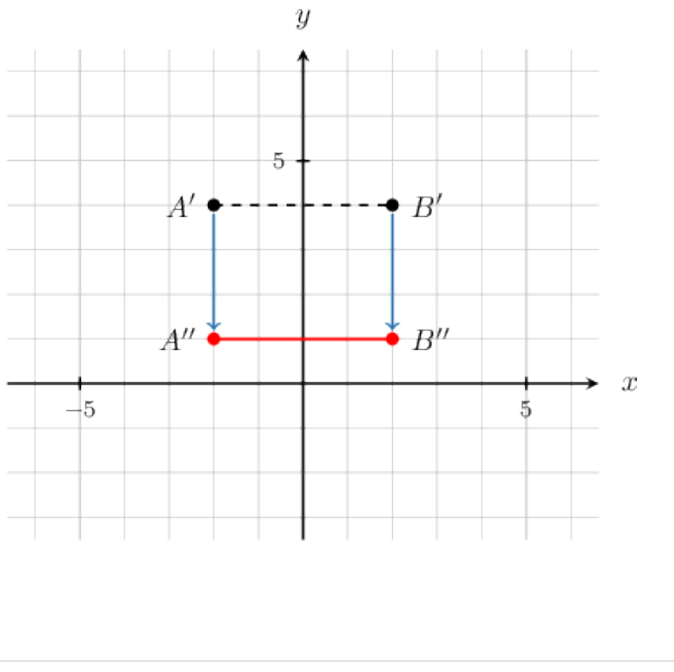


Let's apply the transformations in turn.

- Dilating with center O and scale factor 2 gives the segment $\overline{A'B'}$, shown below:



- Next, we take $\overline{A'B'}$ and shift it by 3 units down. This gives the segment $\overline{A''B''}$:



Example: Determining the Image of a Point Given a Sequence of Transformations

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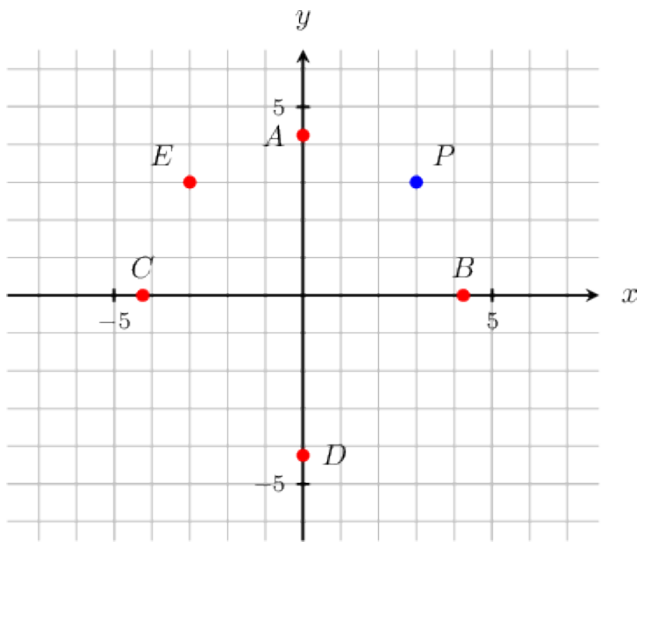
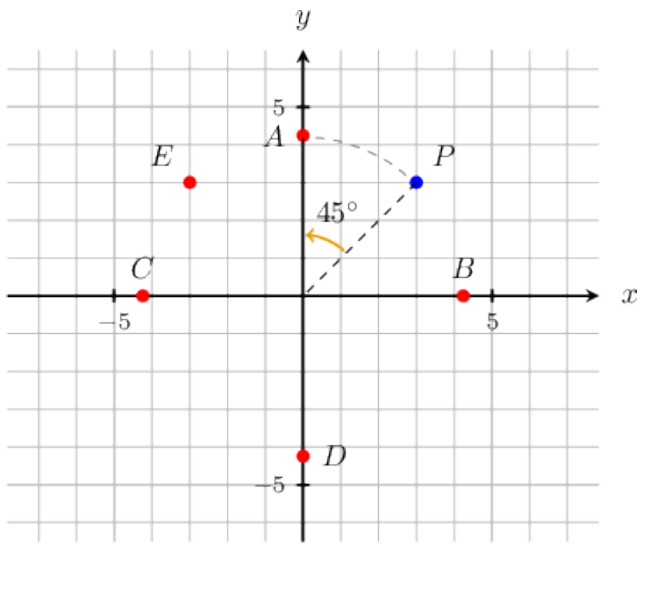
The point P below is rotated by 45° counterclockwise around the origin and then reflected across the x-axis. Which of the given points is the image of P under this sequence of transformations?



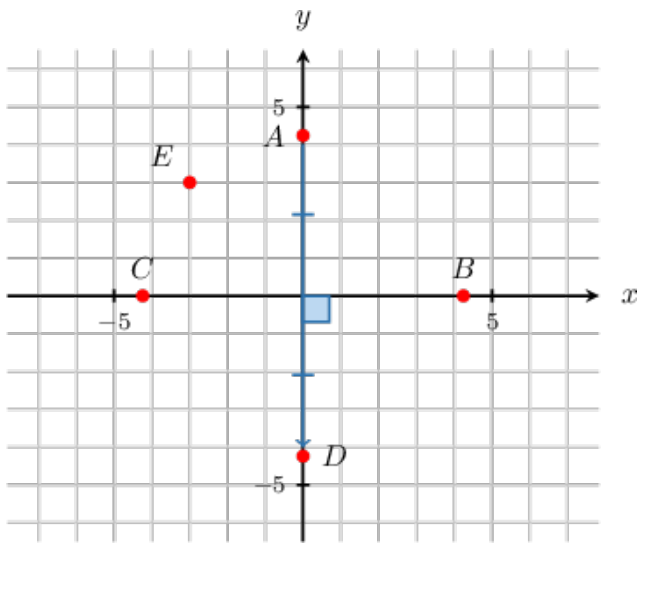
EXPLANATION

Let's apply the given transformations.

- Rotating the point P by 45° counterclockwise around the origin gives the point A, as shown below:



- Reflecting the point A in the x-axis gives the point D, as shown below:



Therefore, the correct answer is D.

KEY PREREQUISITES
Dilations of Figures in the Coordinate Plane
Rotations of Geometric Figures
Reflections of Geometric Figures in the Cartesian Plane

Question 1

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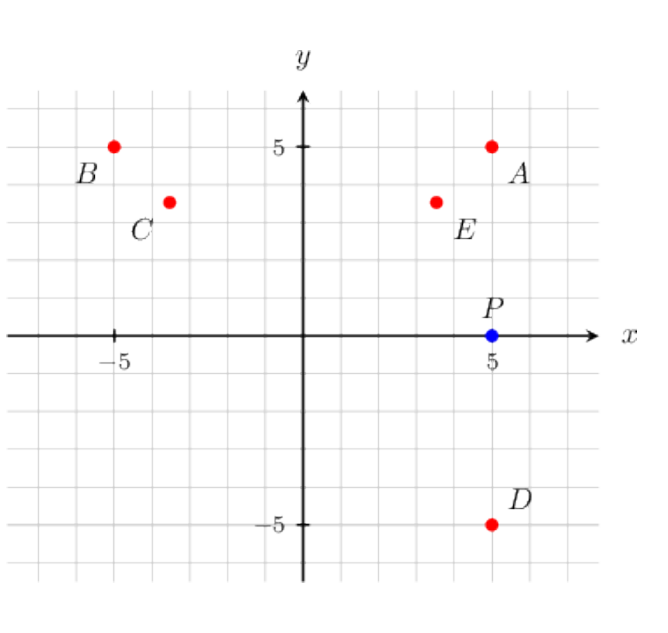

The point P above is rotated by 45° counterclockwise around the origin and then reflected across the y-axis. Which of the given points is the image of P under this sequence of transformations?

- a C
- b B
- c E
- d A
- e D

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Question 2

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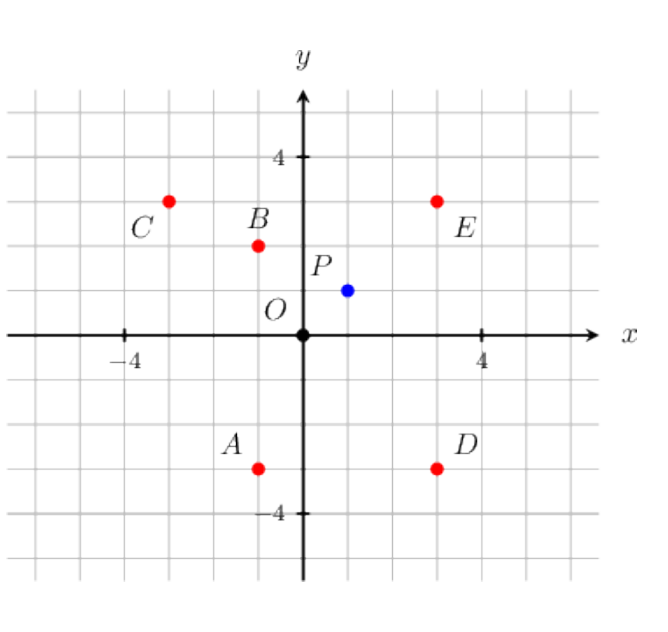

The point P shown above is dilated with center O and scale factor 3 and then reflected across the x-axis. Which of the given points is the image of P under this sequence of transformations?

- a A
- b E
- c B
- d D
- e C

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Example: Applying Multiple Transformations to a Point Using Functions

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The point $Q(4,3)$ is shifted 2 units down, reflected across the y-axis, and then shifted 1 unit to the right. What is the resulting point?

EXPLANATION

Let's recall the functional representations of our three transformations:

- A shift of 2 units down can be represented by the function
$$(x,y)\mapsto(x,y-2).$$
- A reflection in the y-axis can be represented by the function
$$(x,y)\mapsto(-x,y).$$
- A shift of 1 unit to the right can be represented by the function
$$(x,y)\mapsto(x+1,y).$$

Let's apply the given transformations to the point Q using our functions:

First, we apply the shift. The image of the point $(4,3)$ under the shift is
$$(4,3)\mapsto(4,3-2)=(4,1).$$

Then, we apply the reflection. The image of the point $(4,1)$ under the reflection is
$$(4,1)\mapsto(-4,1).$$

Finally, we apply the second shift. The image of the point $(-4,1)$ under the shift is
$$(-4,1)\mapsto(-4+1,1)=(-3,1).$$

Therefore, the resulting point is $(-3,1)$.

KEY PREREQUISITES
Translations of Geometric Figures
Rotating Objects in the Coordinate Plane Using Functions

Question 3

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The point $P(-1,1)$ is shifted 2 units to the left, reflected across the x-axis, and then reflected across the y-axis. What is the resulting point?

- a $(1,3)$
- b $(-1,-1)$
- c $(-1,1)$
- d $(1,2)$
- e $(3,-1)$

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Question 4

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The point $P(10,-2)$ is acted on by a rotation of 90° counterclockwise about O followed by a dilation with center O and scale factor $\frac{1}{2}$. What is the image of P under this transformation?

- a $(1,-5)$
- b $(-1,5)$
- c $(-4,20)$
- d $(4,20)$
- e $(1,5)$

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Example: Determining the Image of a Polygon Under the Action of Multiple Transformations

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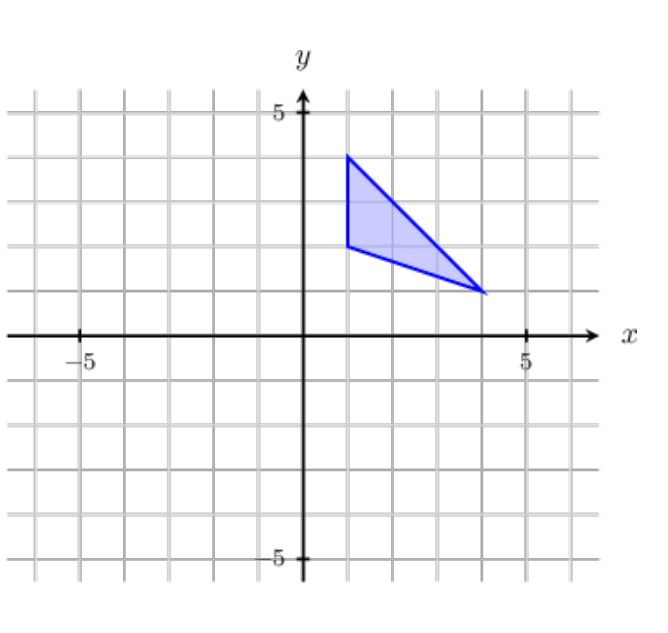
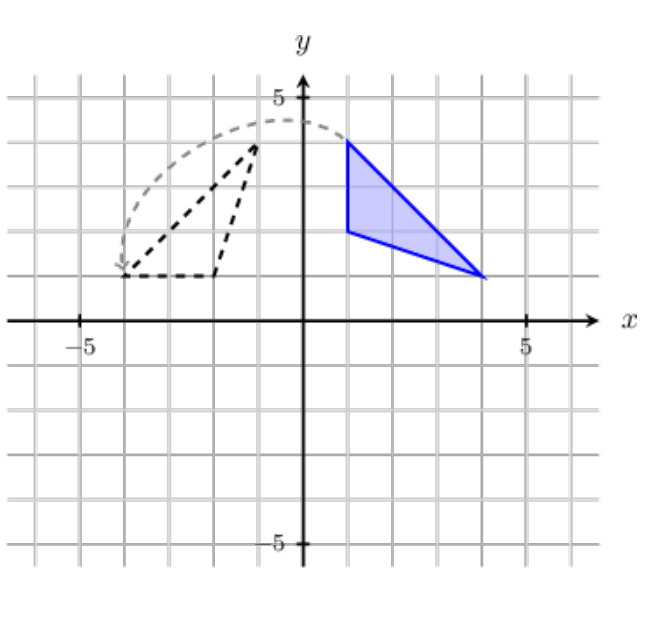


The triangle shown above is rotated by 90° counterclockwise about the origin and then reflected across the x-axis. What is the resulting triangle?

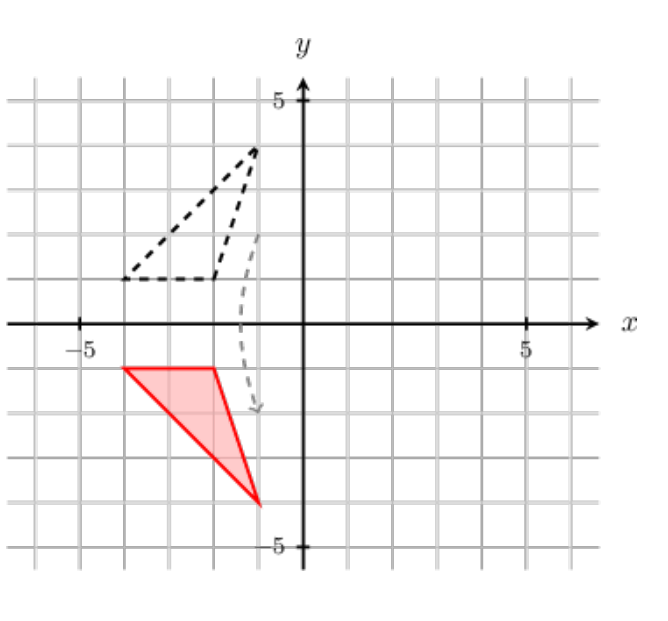
EXPLANATION

Let's apply the given transformation.

- Rotating the triangle by 90° counterclockwise about the origin gives the following:



- Next, we take the last result and reflect it across the x-axis.



Question 5

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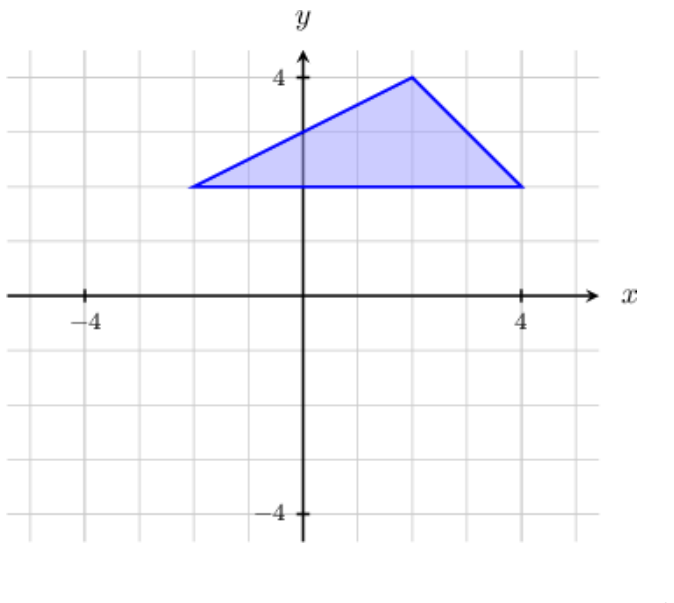
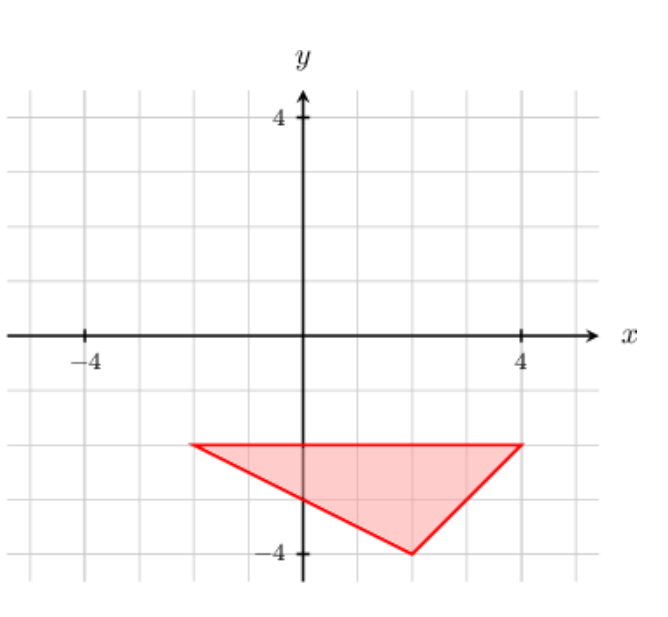
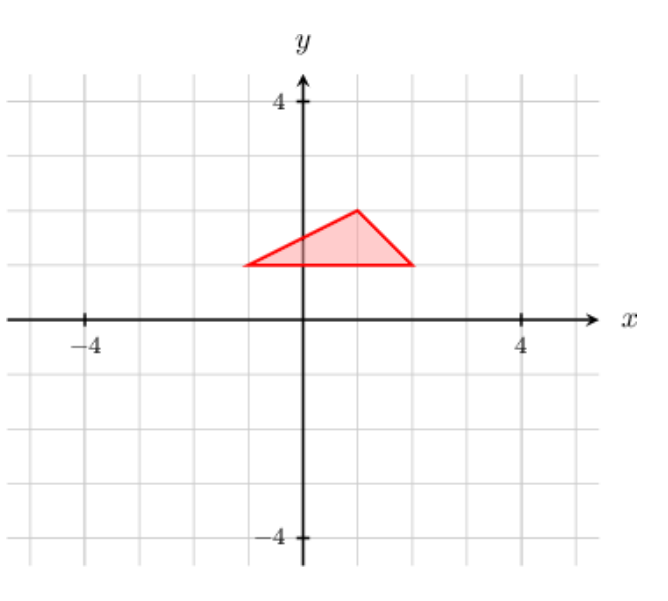
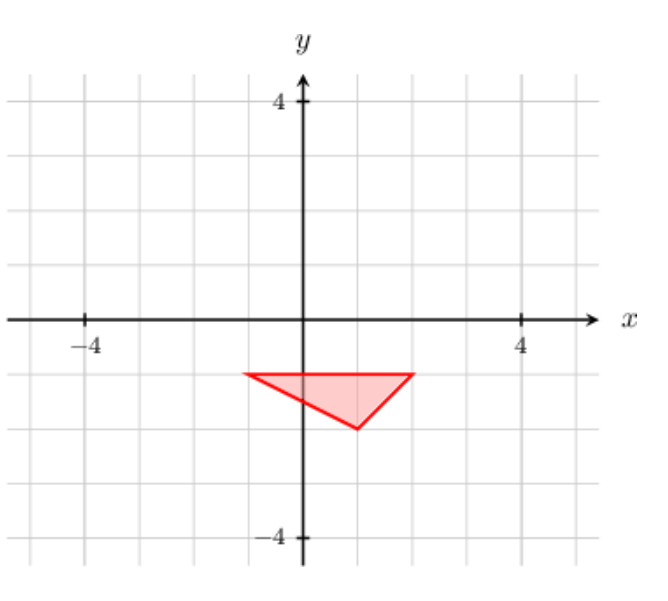
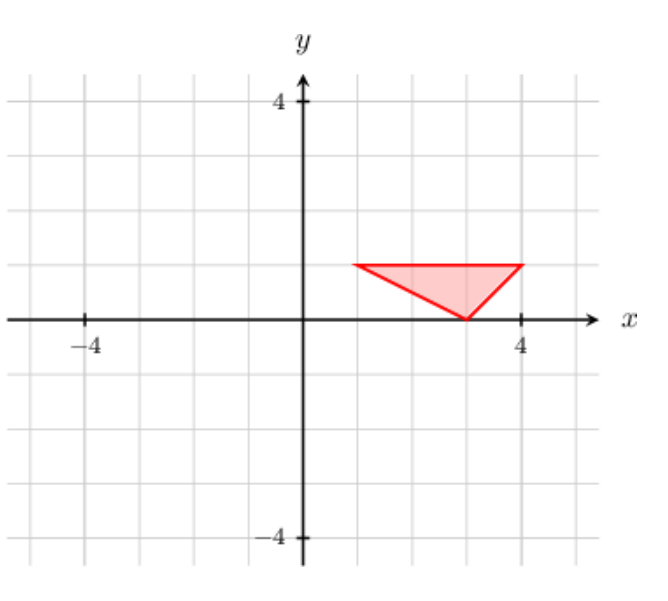


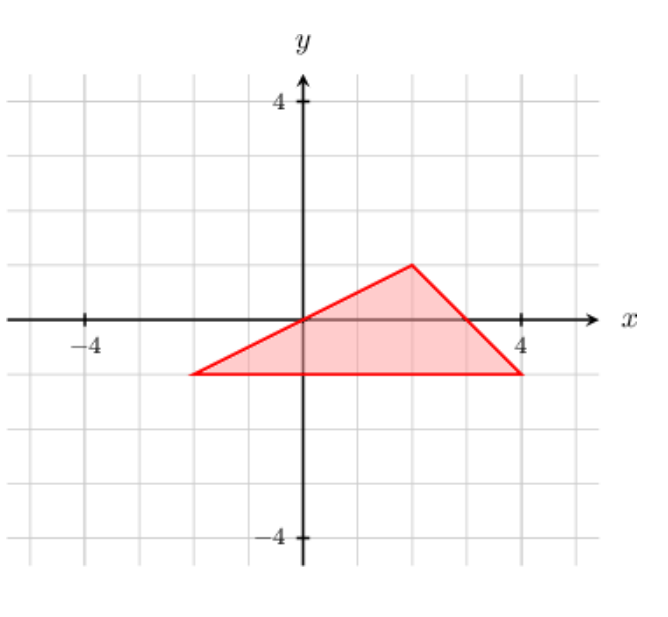
The triangle shown above is transformed subsequently by functions $f_1(x,y)\mapsto\left(\frac{x}{2},\frac{y}{2}\right)$ and $f_2(x,y)\mapsto(x,-y)$. What is the resulting triangle?

a


b


c


d


e


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Question 6

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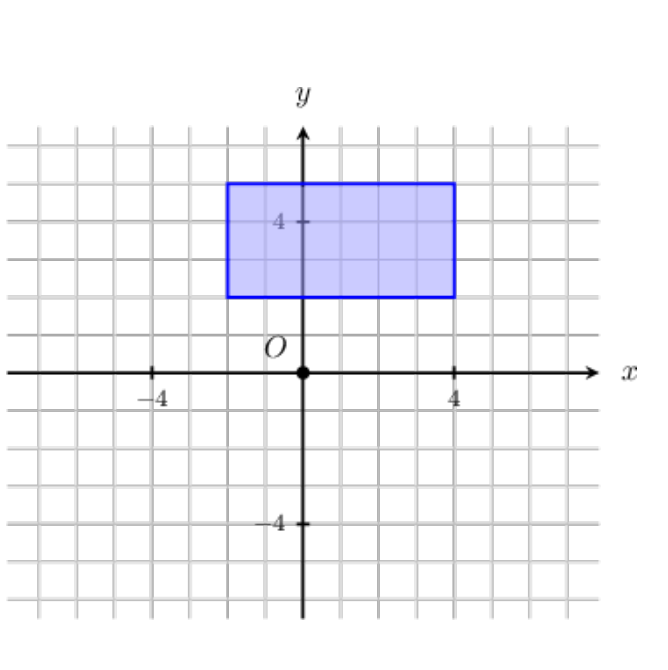
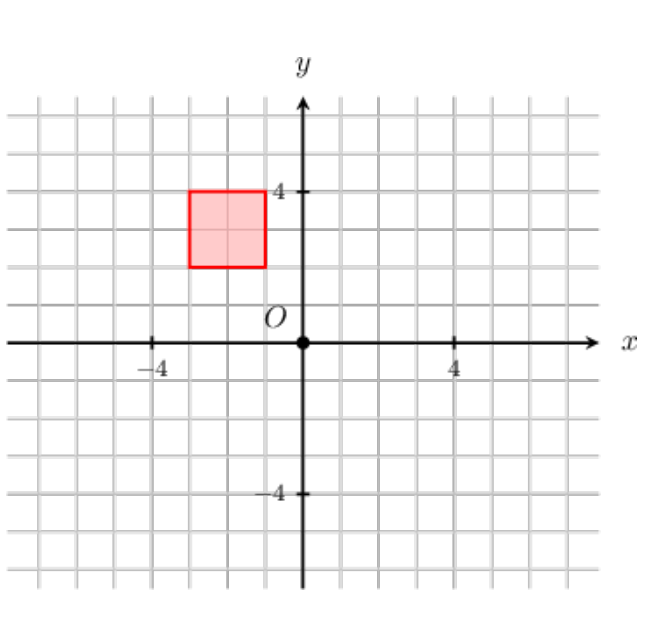


The polygon shown above is reflected in the y-axis and then dilated with center O and scale factor $\frac{1}{2}$. What is the resulting polygon?

a


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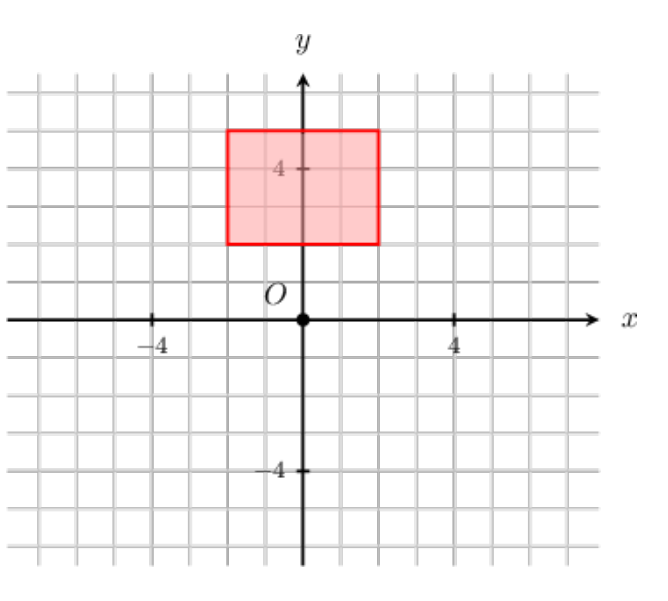
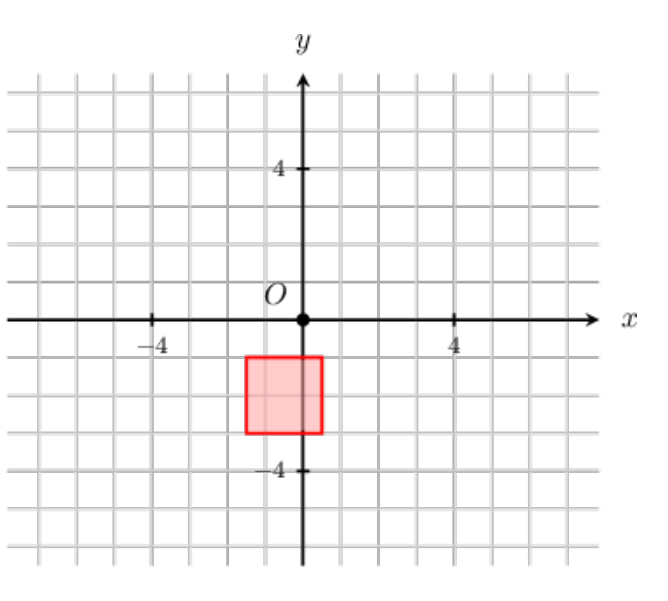
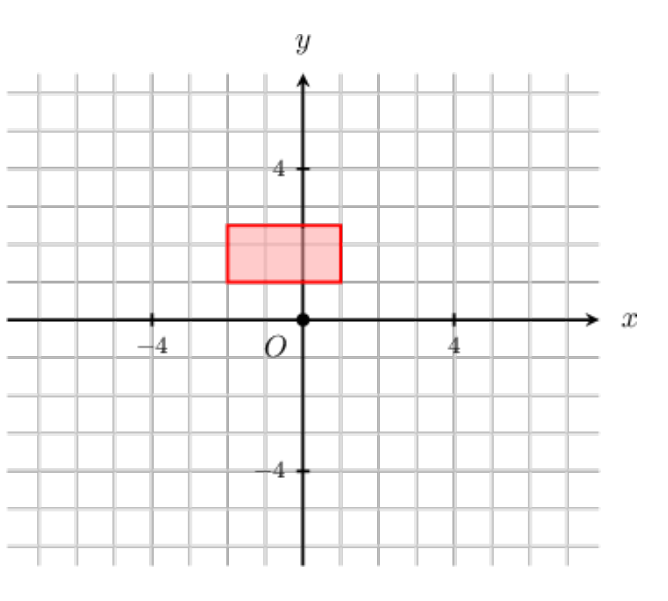
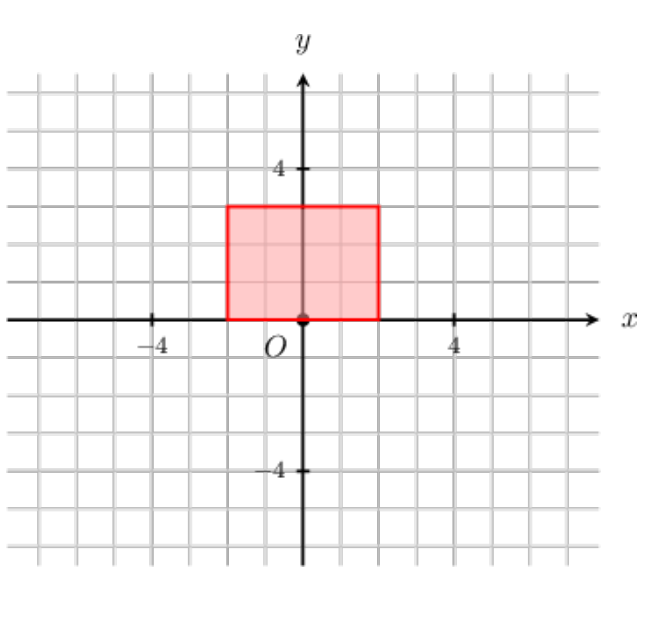


c


d


e


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Example: Determining Parameters Given a Point and Its Image Under Multiple Transformations

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The point $P(5,14)$ is reflected across the line $y=x$ and then stretched according to the function $(x,y)\mapsto(x,ky)$. If the image of P under this sequence of transformations is $(14,25)$, then what is the value of k?

EXPLANATION

Let's apply the given transformations.

A reflection across the line $y=x$ can be represented by
$$(x,y)\mapsto(y,x).$$

Applying this to the point P, we get
$$(5,14)\mapsto(14,5).$$

Now, applying the stretch to $(14,5)$, we get
$$(14,5)\mapsto(14,5k).$$

Since the resulting point is $(14,25)$, by equating the y-coordinates, we obtain
$$5k=25\implies k=5.$$

Therefore, $k=5$.

Question 7

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The point $P(-5,-2)$ is rotated 270° counterclockwise about the origin O and then translated according to the function $(x,y)\mapsto(x+p,y+q)$. If the image of P under this sequence of transformations is $(2,-9)$, then $p+q=$

- a -14
- b -10
- c 4
- d -12
- e 14

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Question 8

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The point $P(5, 6)$ is rotated 90° clockwise about the origin O and then translated according to the function $(x, y) \mapsto (x + k, y + k)$. If the image of P under this sequence of transformations is $(12, 1)$, then $k =$

ⓐ 1

ⓑ 6

ⓒ 12

ⓓ 2

ⓔ 3

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